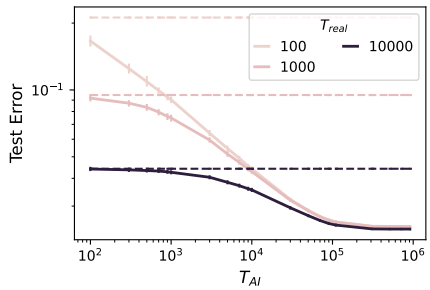

Figure 5. Mixing T_{real} real data with T_{AI} AI data. The dotted lines depict test errors of real data alone. $k = 1,000, \beta = 3/2$.

A first fundamental step is to consider *probabilistic* ground truth labels to replace the deterministic Hutter prediction $i \mapsto y_i$ with a probability distribution $p(j|i)$ on $\mathbb{N}_*$ with power law decay (as in Equation (1)). To account for the fact that the most frequent next token j depends on the preceding token i we model

$$p(j \mid i) \propto \pi_i(j)^{-\beta}, \tag{16}$$

(instead of $j^{-\beta}$), where π_i is a permutation associated to every i providing the order of outputs. To summarize, we think of the data as pairs (i, j), where the distribution of i is governed by some $p(i)$ as in the deterministic Hutter setting, and $p(j|i)$ is given by Equation (16).

This setting can be made *autoregressive* by generating sequences step by step, using the preceding output as the next input. We can think of each successive pair of tokens as of the pairs (i, j) above, with the only difference that the marginal distribution $p(i)$ changes. We thus will make no assumptions on $p(i)$ in what follows (except for a mild technical condition). Proofs can be found in Appendix D.

Remark. The Hutter model abstracts Large Language Models (LLMs) into encapsulated knowledge or skills Michaud et al. (2023), denoted as i. This variable, i, can be interpreted as a composite description comprising multiple components, where the Hutter model highlights the tendency of AI-generated data to exhibit a less pronounced heavy-tailed distribution. In scenarios where AI-generated data encompasses both accurate and erroneous information—such as in predictions of the greatest common divisor—a bias towards incorrect responses is observed. The Hutter++ model extends this framework by incorporating this specific bias. Specifically, we now curate the model to generate AI data with the consistent probability distribution $p(i)$). Errors are represented within the conditional distribution $p(j|i)$, which, in turn, affects the model's performance in later training stages.

4.1. The Hutter++ Algorithm

We now present an extension of the Hutter model (7) which is adapted to bigrams. Let $n_T(i) = \sum_{t=1}^{T} 1[i_t = i]$ be the number times the context i_t appears in the dataset $\mathcal{D}_T$ and $n_T(i, j) = \sum_{t=1}^{T} 1[(i_t, j_t) = (i, j)]$ be the number of times the pair (i, j) appears in the dataset. Note that $n_T(i) \sim Bin(T, p_i)$. As soon as $n_T(i) \geq 1$, define

$$q_T(j \mid i) := n_T(i, j)/n_T(i).$$

This is an empirical version of $p(\cdot \mid i)$ based on an iid sample of size $n_T(i)$. For a theoretical analysis, we shall consider the following test error metric based on total-variation (TV)

$$E_{test} := \sum_i p_i \, \mathbb{E}\left[TV(q_T(\cdot \mid i), p(\cdot \mid i))\right], \tag{17}$$

where $TV(a, b) := \sum_j |a_j - b_j|$ is the total-variation distance and the expectation is over the randomness in q_T. An asset here is that (Berend & Kontorovich, 2012) can be used to control the quantities $\mathbb{E}\left[TV(q_T(\cdot \mid i), p(\cdot \mid i))\right]$. Note that TV is upper-bounded by the square-root of KL-divergence, thanks to *Pinker's inequality.* This gives indication that our results could also apply in the setting of autoregressive models with perplexity loss, like LLMs.

4.2. A Scaling Law for Hutter++

Consider a case of non-deterministic outputs as in Equation 16, where $\pi_1, \pi_2, \ldots$ are functions from $\mathbb{N}_*$ to $\mathbb{N}_*$.

Theorem 4.1. *Suppose $\beta \in (1, \infty) \setminus \{2\}$ and set $c := \min(1 - 1/\beta, 1/2)$. If $\sum_i p_i^{1-c} < \infty$, then $E_{test} \lesssim T^{-c}$. Moreover, if $\beta \in (1, 2)$ and the mappings $\pi_1, \pi_2, \ldots$ are permutations, then $E_{test} \asymp T^{-c}$.*

Thus, the proposed Hutter++ algorithm induces exactly the same scaling law as the classical setup (Hutter, 2021) !

4.3. Model Collapse in Probabilistic Setting

We now return to our main problem, understanding model collapse in the probabilistic setting and consider the Hutter++ presented above. Thus, suppose the learner only has access to at most a dataset of size T containing the kth head of the conditional distribution $p(\cdot \mid i)$. That is, sampled from: $i \sim p$, $j \sim p(j \mid i)1[j \leq k]$ (normalized appropriately), where $p(\cdot \mid i)$ is as in Equation (16).

Theorem 4.2. *(A) If $\beta \in (1, \infty) \setminus \{2\}$ and $\sum_i p_i^{1-c} < \infty$ where $c := \min(1 - 1/\beta, 1/2)$ as before, then $E_{test} \lesssim T^{-c} + k^{-\beta c}$.*

(B) Furthermore, if the mappings $\pi_1, \pi_2, \ldots$ are permutations and $\sum_i p_i^{1-c} < \infty$, then $E_{test} \asymp T^{-c} + k^{-\beta c}$.

Autoregressive Bigrams. Similarly these results hold for autoregressive bigram model, where $p(i_1, i_2, \ldots, i_L) = p(i_1) \prod_{\ell=1}^{L-1} p(i_{\ell+1} \mid i_\ell)$, and each $p(j \mid i)$ is as in (16). The result is empirically confirmed in Figure 11 in Appendix B.

Multiple Generations. The mechanics of the proof of Theorem 2.4 apply in this setting. See Figure 12 in Appendix B

illustrating that Equation (13) keeps holding for probabilistic data distributions.

Grokking for Mixtures. Technically speaking, this grokking phenomenon only holds for models with deterministic ground truth labels, like the Hutter LLM and the limited capacity associative memory model. For the *probabilistic* setting of bigrams (or text LLMs) the theorem cannot hold in its pure form, because if we train on a mixture of two distributions (clean and synthetic) but test only on the clean distribution, the distance between these two distributions will always be a lower bound on the test error. However, we can see that remnants of a "smoothed" grokking-law persist in the form of an S-shaped scaling (see Figure 19).

5. Capacity-Limited Memory Models: A Triplet Scaling Law

We now turn to a finite-memory extension of the Hutter LLM, which allows to model *capacity*. We look into a simple associative memory model (Cabannes et al., 2023):

$$\begin{aligned} f_T(i) &:= \arg\max_y H_T(i,y), \text{ where} \\ H_T(i,y) &:= e_i^\top M_T u_y, \\ M_T &:= \sum_i q_T(i) e_i u_{f_\star(i)}^\top \in \mathbb{R}^{d\times d}. \end{aligned} \tag{18}$$

This is a transformer-like finite-memory extension of the infinite-memory model in (Hutter, 2021). The integer $d \geq 1$ then plays the role of the "capacity" of the resulting model. Here, $f_\star : [N] \to [m]$ is an unknown function, for example, reduction modulo m, i.e $f_\star(i) := ((i-1) \bmod m)+1$; $q_T = q(\mathcal{D}_T)$ is probability distribution on $[N]$ which encodes an arbitrary learner, estimated using and iid sample $\mathcal{D}_t = \{(i_t, y_t) \mid t \in [T]\}$ of size T collected from a probability distribution on $[N] \times [m]$, of the form

$$i \sim p = \text{Zipf}(\beta), \quad y = f_\star(i). \tag{19}$$

The embedding vectors $e_1, e_2, \ldots e_N$ and $u_1, u_2, \ldots, u_m$ are a system of unit-vectors in $\mathbb{R}^d$, constructed so that the matrix $\mathbb{R}^{d\times d}$ remembers the input/output pairs (i,j) it has seen, i.e $e_i^\top M u_{f_\star(i)} \approx q_T(i)$ if $(i, f_\star(i)) \in \mathcal{D}_T$. The weights $q_T(i)$ ensure that different memories are memorized faster than others.

Cabannes et al. (2023) proposed that iid random embeddings from the uniform distribution on the unit-sphere in $\mathbb{R}^d$ be used. In this setting, for different choices of q, the following general scaling law was established

$$E_{test} \asymp T^{-(1-1/\beta)} + d^{-c_q}, \tag{20}$$

where the exponent $c_q \in (0,\infty)$ depends on β and the algorithm q. For example, when q encodes the counting measure $q_T(i) := n_T(i)/T$ (reminiscent of SGD), it was shown that $c_q = (1-1/\beta)/2 \in (0, 1/2)$. Another algorithm $q_T(i) := 1[n_T(i) \geq 1]/\sum_\ell 1[n_T(\ell) \geq 1]$ (remniscent of ADAM) was proposed which attains a optimal error rate (over all algorithms based on random embeddings) with $c_q = \beta - 1$.

In the context of model collapse which is the main focus of this manuscript, we have the following.

Theorem 5.1 (Triplet Scaling Law)**.** *For all the algorithms q considered in (Cabannes et al., 2023), one has the following triplet scaling law w.r.t sample size T, embedding dimension d, and frequency cutoff k,*

$$E_{test} \asymp T^{-(1-1/\beta)} + d^{-c_q} + k^{-(\beta-1)}. \tag{21}$$

This result is empirically confirmed in Figure 21 and proved in Appendix E. It gives rise to similarly tapered-off scaling curves for synthetic data, as in the simpler models. The proofs for loss of scaling across generations in Section 2 and grokking phenomena in Section 3, carry over to this model as well, demonstrating their universality.

6. Experiments

In this section we present our experimental results to demonstrate evidence of various predictions we have made theoretically. We showcase four scenarios of increasing level of complexity: an empirical Hutter++ model, autoregressive bigram models with *perplexity loss*, an arithmetic transformer to predict the GCD of two integers (Charton, 2023) and a large-scale LLM, Llama2-7B (Touvron et al., 2023), trained on a large data corpus (`Wikidata-103`).

In our theoretical analysis, motivated by empirical observations (see Figure 2) or by the effect of finite data sampling bias on heavy-tailed data, we have assumed that generated data follows patterns of either a tail cutoff or tail narrowing. In our subsequent experiments, we depart from theoretical assumptions on tail-cutting/narrowing to allow the widely deployed top-p^{inf} selection or temperature scaling mechanisms to give possibly intermingled effects on the generated data distribution.

Empirical Hutter++ Model. In Figure 6, we use an initial model that is trained on $T_0 = 100,000$ samples from the original distribution. For the Gen 1 line, the data are all generated from this initial model. From Gen 2 onwards, models are iteratively trained on data produced by the most performant model of the preceding generation, effectively eliminating the possibility that model collapse results from inadequate sampling. For Gen 1, a notable degradation in data scalability is observed, alongside a rapid decline in model performance across generations. These observations not only validate our theoretical result but also reaffirm our assumptions. A similar pattern is evident with temperature scaling, as shown in Figure 16.

Autoregressive Bigram Models with Perplexity Loss. We

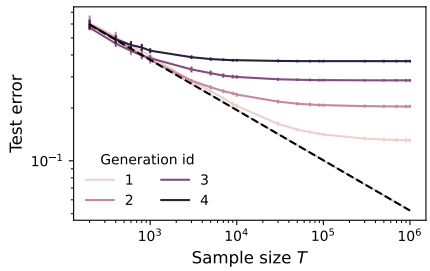

Figure 6. **Hutter++ on Bigram with limited data and top-p^{inf}.** The initial model is trained on $T_0 = 100,000$ samples. It generates T samples for Gen 1. Starting from Gen 2 models are trained on data generated by the most powerful model from the previous generation. Top-p^{inf} =0.95 cutting and $\beta = 3/2$.

move one step further towards "real" LLMs to investigate autoregressive bigram models. The dataset now comprises sequentially generated integers, adhering to Equation (16), with the model trained on all tokens. We use the averaged perplexity score of the test set as the test error metric. Our study encompasses a range of effects—such as top-p^{inf} inference, temperature scaling, limited real data, and training on progressively larger AI datasets. Consistent with the findings in Section 4, we observe the same patterns of scaling loss and progressive model collapse across generations. Relevant figures are provided in Appendix F.

Transformers Learning the GCD. Our first illustration of our theory "in the wild" is for sequence-to-sequence transformer models for an arithmetic task: predicting the greatest common divisor (GCD) of two integers, encoded as sequences of digits in some base B, following Charton (2023). This setup is a perfect intermediate step between our toy models and large scale LLMs; it uses the transformer architecture and training algorithms on sizeable models, while the underlying data has a deterministic nature. Over the course of training the model progressively learns new GCDs and with them also their products with already learned GCDs. We can thus view each such learned group, usually learned in "bursts", as a new skill. For the purpose of this experiment, we use this model after $300M$ samples as the generator of AI-data. In Figure 4 we validate the predicted scaling law for a single generation and observe 'un-learning' of skills when training exclusively with generated data, as well as a grokking effect when training on mixtures. See Appendix G for details and more figures.

Experiments on LLMs. We finetune Llama2 with LoRA, generating synthetic AI data for the next finetuning iteration. Inspired by the setup in Shumailov et al. (2023), we use `Wikidata-103`, partitioned into approximately 2.2 million sequences of 128 tokens. AI data is generated through prompt completion, using the first 96 tokens from the original sequences as prompts. The model is trained only on the last 32 tokens to preclude information leakage. The evaluations are conducted exclusively on the same 32 tokens. We use top-p^{inf} =0.9 and temperature τ =0.9 across all generations. The results, depicted in Figure 4 (left), illustrate a scaling law decay over several generations. The first generated dataset still contains useful but limited information; the utility of the second generation's data markedly diminishes. These phenomena corroborate the anticipated loss of scaling law and model collapse, further indicating that model collapse is even more pronounced here, highlighting the challenges in training next generation LLMs. More details and results in Appendix H.

Moreover, we conduct experiments to investigate mixing a proportion of real data with AI-generated data. Figure 7 demonstrates the effect of blending a random 2% of original data with AI data across all fine-tuning phases. It significantly mitigates model collapse, with the emergence of a grokking curve as predicted in Theorem 3.2.

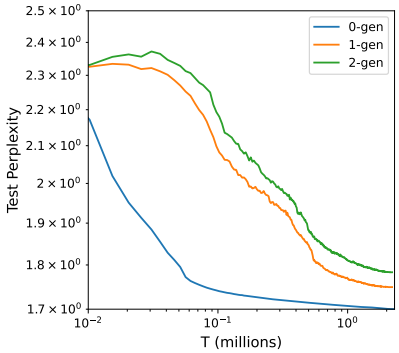

Figure 7. **Mixing Llama Generated Data with Original Data** Based on Figure 4 left, we mix generated data with original data, with ratio 98 to 2. Note how the mixing curve validates our predicted curve of the grokking phenomenon as in Figure 3

7. Conclusion

In the advent of the "synthetic data age", our work signals the end of current neural scaling laws and opens the door to a puzzling array of new phenomena in a world where the training corpora are enriched with AI generated data. We demonstrate that scaling laws cease to persist; test error tapers off due to altered, less heavy tailed, data distributions. Yet, new opportunities arise from careful mixture and data curation, as we have shown, with interesting effects at the interplay of clean and synthesized data. We must recognize new learning plateaus and, for instance, adjust to changed learning curves from blending clean and synthetic data to unintended early stopping. A notable feature of our work is that our theory is *effective* - we observe the predicted phenomena for relevant large models in two different settings.

Our contributions call for a more responsible, or "collapse-aware", proliferation of synthesized data. Scale is *not* all you need: more work on watermarking for synthetic data is needed, to make it more distinguishable from original human-annotated data. "Real" data will become an even more valuable resource in the future, as we are ushering in the "beyond scaling" era.